Chapter-4

Random Variable and Probability distribution.

Q) In a continuous distribution whose pdf is given by

$f(x) = Ax(2-x)$, $0 \leq x \leq 2$.

i> Find the constant value of A.

ii> $P(X \leq 1)$

iii> $P(x \leq 1/2)$

iv> $P(1/2 \leq x \leq 3/2)$

v> $P(x \geq 4/3)$

vi> Expected value of x.

Sol Here,

$$f(x) = Ax(2-x) ; \quad 0 \leq x \leq 2$$

i> By definition of p.d.f,

$$\int_x f(x)\,dx = 1$$

$$\Rightarrow \int_0^2 Ax(2-x)\,dx = 1$$

$$\Rightarrow A\int_0^2 (2x-x^2)\,dx = 1$$

$$\Rightarrow A\left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_0^2 = 1$$

$$\Rightarrow A \times \left(4 - \frac{8}{3}\right) = 1$$

$$\Rightarrow A \times \left(\frac{4}{3}\right) = 1$$

$$\therefore A = 3/4$$

$$\therefore f(x) = 3/4\, x(2-x) ; \quad 0 \leq x \leq 2$$

$\therefore$ The constant value of A is 3/4. Ans i> $P(x \leq 1) = \int_0^1 f(x)\,dx = \int_0^1 \frac{3}{4} x(2-x)\,dx = 0.5$ Ans

ii> $P(x \leq \frac{1}{2}) = \int_0^{1/2} f(x)\,dx = \int_0^{1/2} \frac{3}{4} x(2-x)\,dx = 0.15625$ Ans

iii> $P(\frac{1}{2} \leq x \leq \frac{3}{2}) = \int_{1/2}^{3/2} f(x)\,dx = \int_{1/2}^{3/2} \frac{3}{4} x(2-x)\,dx = 0.6875$ Ans

~~iv> P(1/2 ≤~~

v> $E(X) = \int_0^2 x f(x)\,dx = \int_0^2 3x^2(2-x)\,dx = 1$ Ans

(8) A continuous random value of x has the following pdf $f(x) = Ax^2,\ 0 \leq x \leq 1$. Find:

i> the value of A.

ii> Expected value of x or mean.

iii> variance of x.

iv> Calculate expectation of $E(3x+2)$, $V(5x+3)$, $E(5x^2)$.

Sol: Here,

$f(x) = Ax^2,\ 0 \leq x \leq 1$

i> By the definition of pdf,

$$\int_x f(x)\,dx = 1$$

$$\Rightarrow \int_0^1 Ax^2\,dx = 1$$

$$\Rightarrow A \times 0.33 = 1$$

$$\therefore A = \frac{1}{0.33} = 3 \text{ Ans}$$

. The value of A is 3.

$\therefore f(x) = 3x^2 \quad ; \quad 0 \le x \le 1$

ii> Expected value of x, $E(x) = \int_x x f(x)\, dx = \int_0^1 x \cdot 3x^2 dx$

$= 0.75$ Ans.

iii> Variance of x ; $E(x^2) = \int_x x^2 f(x)\, dx = \int_0^1 x^2 \cdot 3x^2 dx = 0.6$ Ans

$\therefore V(X) = E(x^2) - (E(X))^2$

$= 0.6 - (0.75)^2$

$= 0.0375$ Ans

iv>

$E(3X+2) = 3E(X) + 2 = 3 \times 0.75 + 2 = 4.25$ Ans

$V(5x+3) = 5^2 V(X) + 0 = 25 \times 0.0375 = 0.9375$ Ans

$E(5X^2) = 5E(X^2) = 5 \times 0.6 = 3$ Ans

4. Let x be a random variable with p.d.f

$$f(x) = \begin{cases} ax & ; 0 \le x \le 1 \\ a & ; 1 \le x \le 2 \\ -ax+3a & ; 2 \le x \le 3 \\ 0 & ; \text{otherwise} \end{cases}$$

Find:

i> the value of a.

ii> $P(X \le 1.5)$

iii> Expectation of x.

iv> variance of x.

v> $E(3X+5)$, $V(3x+1)$.

Sol: Here,

$$f(x) = \begin{cases} ax & ; 0 \le x \le 1 \\ a & ; 1 \le x \le 2 \\ -ax+3a & ; 2 \le x \le 3 \\ 0 & ; \text{otherwise} \end{cases}$$

i> By the definition of p.d.f,

$$\int_x f(x)\,dx = 1$$

$$\Rightarrow \int_0^3 f(x)\,dx = 1$$

$$\Rightarrow \int_0^1 ax\,dx + \int_1^2 a\,dx + \int_2^3 (-ax+3a)\,dx = 1$$

$$\Rightarrow \left[\frac{ax^2}{2}\right]_0^1 + \left[ax\right]_1^2 + \left[-\frac{ax^2}{2} + 3ax\right]_2^3 = 1$$

$\Rightarrow \frac{a}{2} + 2a - a + \left[-\frac{9a}{2} + \frac{4a}{2} + 9a - 6a \right] = 1$

$\Rightarrow \frac{a}{2} - \frac{9a}{2} + \frac{4a}{2} + 2a - a + 3a = 1$

$\Rightarrow \frac{-4a}{2} + 4a = 1$

$\Rightarrow -2a + 4a = 1$

$\Rightarrow 2a = 1$

$\therefore a = 1/2$

$\therefore$ The value of a is $1/2$.

Hence,

$$f(x) = \begin{cases} \frac{1}{2}x & ; \quad 0 \leq x \leq 1 \\ 1/2 & ; \quad 1 \leq x \leq 2 \\ -\frac{1}{2}x + 3/2 & ; \quad 2 \leq x \leq 3 \\ 0 & ; \quad \text{otherwise} \end{cases}$$

ii> $P(X \leq 1.5) = \int_0^{1.5} f(x)\,dx = \int_0^1 \frac{1}{2} x\,dx + \int_1^{1.5} a\,dx = 0.5$ Ans

iii> Expectation of x: $E(X) = \int_x x f(x^2)\,dx$

$= \int_0^1 \frac{1}{2} x^2\,dx + \int_1^2 \frac{1}{2}x\,dx + \int_2^3 x\left(-\frac{1}{2}x + \frac{3}{2}\right)dx$

$= \frac{3}{2} = 1.5$ Ans

iv) Variance of X:

$$E(X^2) = \int_x x^2 f(x)\, dx$$

$$= \int_0^1 x^2 \frac{1}{2} x\, dx + \int_1^2 \frac{x^2}{2}\, dx + \int_2^3 \left(\frac{-x^3}{2} + \frac{3x^2}{2}\right) dx$$

$$=$$

$$\therefore V(X) = E(X^2) - [E(X)]^2$$

$$= 2.67 - (1.5)^2$$

$$= 0.42 \text{ Ans}$$

$P(X=x) = {}^nC_x \, p^x q^{n-x}$ $\left(\because p+q=1\right)$

$np = \text{mean} = E(X)$; $p = ---$

$npq = \text{variance}$; $q = 1-p$

Binomial distribution

1. If a coin is tossed 10 times simultaneously. What is the probability of getting

i> five heads ?

ii> no heads ?

iii> at least one heads ?

iv> at most three heads ?

v> at most nine heads?

sol

Let X be a random variable that denotes the no. of heads.

no. of trials, $n = 10$

prob. of head, $p = 1/2$

prob. of tail, $q = 1 - 1/2 = 1/2$

no. of heads $= x$

we have,

$$P(X=x) = {}^nC_x \, p^x q^{n-x}$$

$$= {}^{10}C_x (1/2)^x (1/2)^{10-x}$$

i> $P(\text{five heads}) = P(X=5)$

$$= {}^{10}C_5 \, (1/2)^5 (1/2)^{10-5}$$

$$= \frac{63}{256} \quad \text{Ans}$$

ii> $P(\text{no heads}) = P(X=0)$

$$= {}^{10}C_0 \, (1/2)^0 (1/2)^{10-0}$$

$$= \frac{1}{1024} \quad \text{Ans}$$

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iii> $P(\text{at least one head}) = P(X \geq 1)$

$= 1 - P(X < 1)$

$= 1 - P(X=0)$

$= 1 - \frac{1}{1024}$

$= \frac{1023}{1024}$ Ans

iv> $P(\text{at most three heads}) = P(X \leq 3)$

$= P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$= {}^{10}C_0 (1/2)^0 (1/2)^{10-0} + {}^{10}C_1 (1/2)^1 (1/2)^{10-1}$
$+ {}^{10}C_2 (1/2)^2 (1/2)^{10-2} + {}^{10}C_3 (1/2)^1 (1/2)^{10-3}$

$= \frac{11}{64}$ Ans

v> $P(\text{at most three tails}) = P(7 \text{ heads})$

$= P(X = 7)$

$= {}^{10}C_7 (1/2)^7 (1/2)^{10-7}$

$= \frac{15}{28}$ Ans

vi> $P(\text{at most 9 heads}) = P(X \leq 9)$

$= 1 - P(X > 9)$

$= 1 - P(X=10)$

$= 1 - {}^{10}C_{10} (1/2)^{10} (1/2)^{10-10}$

$= \frac{1023}{1024}$ Ans

Q) A multiple choice test has 10 questions. E has 4 options. What is the probability of getting

i) two questions?

ii) at least one questions?

iii) at most two questions?

iv) at most three questions are answered correctly by guessing only?

Sol

Let x be a random variable that denotes the no. of correct answers.

no. of questions, $n = 10$

prob. of correct answer, $p = 1/4$

prob. of wrong answer, $q = 1-p = 1-1/4 = 3/4$

no. of correct answer $= x$

we know,

$$P(X=x) = {}^nC_x\, p^x\, q^{n-x}$$
$$= {}^{10}C_x (1/4)^x \cdot (3/4)^{10-x}$$

i) P(two question answered correctly) : $P(X=2)$

$$= {}^{10}C_2\ (1/4)^2\ (3/4)^{10-2} = 0.281 \text{ Ans}$$

ii) P(at least one question) : $P(X \geq 1) = \cancel{P(X=0)+P(X=1)}$

$$= 1 - P(X<1)$$
$$= 1 - P(X=0)$$
$$= 1 - {}^{10}C_0 (1/2)^0 (3/4)^{10-0}$$
$$= 0.94 \text{ Ans}$$

iii> P(at most two questions answered correctly):

$P(X \le 2) = P(X=0) + P(X=1) + P(X=2)$

$= {}^{10}C_0 (1/4)^0 (3/4)^{10-0} + {}^{10}C_1 (1/4)^1 (3/4)^{10-1} + {}^{10}C_2 (1/4)^2 (3/4)^{10-2}$

$= 0.52$ Ans

iv> P(at least three questions answered correctly):

$P(X \ge 3) = 1 - P(X < 3)$

$= 1 - \cancel{P(X=0) - P(X=1)}\ P(X \le 2)$

$= 1 - 0.52$

$= 0.48$ Ans

(13) Suppose the mortality rate for a certain disease is 0.1. Suppose 10 people in a community contact with the disease. What is the probability that at least 7 people will survive?

Sol

Let X be a random variable that denotes the no. of people survived.

no. of people, $n = 10$

prob. of survival, $p = 1 - 0.1 = 0.9$

prob. of dead, $q = 0.1$

no. of survival, $x = 7$

we know,

$P(X=x) = {}^{n}C_x\, p^x\, q^{n-x}$

$= {}^{10}C_x (0.9)^x (0.1)^{10-x}$

P(atleast 7 people will survive) : $P(X \geq 7)$

$= \cancel{1 - P(X < 7)}$

$= P(X=7) + P(X=8) + P(X=9) + P(X=10)$

$= {}^{10}C_7 (0.9)^7 (0.1)^{10-7} + {}^{10}C_8 (0.9)^8 (0.1)^{10-8} + {}^{10}C_9 (0.9)^9 (0.1)^{10-9} + {}^{10}C_{10} (0.9)^{10} (0.1)^{10-10}$

$= 0.9$ Ans

④ For a binomial distribution of random variable of X, mean = 3 and variance = 2. Find the probability, $P(X \leq 2)$.

Sol: Here,

mean, $np = 3$

variance, $npq = 2$

$P(X \leq 2) = ?$

we know,

$np = 3$

$\therefore n = 3/p$

$npq = 2$

$\Rightarrow 3q = 2$

$\therefore q = 2/3$

$\therefore p = 1 - 2/3 = 1/3$

$\therefore n = 3/p = 3/\frac{1}{3} = 9$

Now,

$P(X \le 2) = P(X=0) + P(X=1) + P(X=2)$

$= {}^{10}C_0 (1/3)^0 (2/3)^{10-0} + {}^{10}C_1 (1/3)^1 (2/3)^{10-1}$

$+ {}^{10}C_2 (1/3)^2 (2/3)^{10-2}$

$= 0.377$ Ans

(6) A coin is tossed until a head appears. What is the ~~probability~~ expectation of no. of tosses required?

sol:

Let X be a r.v. that denotes the no. of tosses required to the ~~th~~ first head. Then we have,

Expectation of X, $E(X) = \sum_{i=1}^{\infty} x_i P(x_i)$

Now,

Events	x_i	$P(x_i)$
H	1	$1/2$
TH	2	$1/2 \times 1/2 = 1/4$
TTH	3	$1/2 \times 1/2 \times 1/2 = 1/8$
TTTH	4	$1/2 \times 1/2 \times 1/2 \times 1/2 = 1/16$
TTTTH	5	$1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 = 1/32$
⋮	⋮	⋮
⋮	⋮	⋮

Now,

$$E(X) = \sum_{i=1}^{\infty} x_i P(x_i) = 1\times\frac{1}{2} + 2\times\frac{1}{4} + 3\times\frac{1}{8} + 4\times\frac{1}{16} + 5\times\frac{1}{32} + \cdots$$

$$\therefore E(X) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots$$

Let $E(X) = S = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots$ —— (1)

Since this series is Arithmetic Geometric series. so multiply (1) by common ratio of G.S. $(r) = 1/2$.

$$\Rightarrow \frac{1}{2}S = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \frac{5}{64} + \cdots \text{ —— (2)}$$

from (2) & (3), we get;

$$S - \frac{1}{2}S = \left(\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots\right)$$

$$- \left(\frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \frac{5}{64} + \cdots\right)$$

$$\Rightarrow \frac{1}{2}S = \frac{1}{2} + \left(\frac{2}{4} - \frac{1}{4} + \frac{3}{8} - \frac{2}{8} + \frac{4}{16} - \frac{3}{16} + \frac{5}{32} - \frac{4}{32} + \cdots\right)$$

$$\Rightarrow \frac{1}{2}S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \cdots$$

$$\Rightarrow \frac{1}{2}S = \frac{1/2}{1 - 1/2} \qquad \left(\because S_\infty = \frac{a}{1-ar}\right)$$

$$\Rightarrow S = 2$$

i.e. $E(X) = 2$

Thus, the expectation of no. of tosses required.

Imp 2023-Spring

(6.) A coin is tossed until a head appears. What is the expectation of no. of tosses required?

Sol

Let X be a random variable that denotes the no. of tosses required.

Expectation of X, $E(X) = \sum_{i=1}^{\infty} x_i p(x_i)$

Now,

Events	x_i	$p(x_i)$
H	1	$\frac{1}{2}$
TH	2	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
TTH	3	$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$
TTTH	4	$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$
TTTTH	5	$\frac{1}{2} \times \frac{1}{16} = \frac{1}{32}$
⋮	⋮	⋮

$$\therefore E(X) = \sum_{i=1}^{\infty} x_i (P x_i)$$

$$= 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + \frac{4}{16} + 5 \times \frac{1}{32} + \cdots$$

Let $S = E(X) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots \text{———} (1)$

Since this series is Arithmetic-Geometric series, so multiply (1) by common ratio ($r = \frac{1}{2}$).

$\Rightarrow \frac{1}{2}S = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \frac{5}{32} + \cdots\cdots$ —— (2)

from (1) & (2), we get;-

$S - \frac{1}{2}S = \left(\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots\cdots\right) - \left(\frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \frac{5}{64} + \cdots\cdots\right)$

$\frac{S}{2} = \frac{1}{2} + \left(\frac{2}{4} - \frac{1}{4} + \frac{3}{8} - \frac{2}{8} + \frac{4}{16} - \frac{3}{16} + \frac{5}{32} - \frac{4}{32} + \cdots\cdots\right)$

$\Rightarrow \frac{S}{2} = \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \cdots\cdots\right)$

$\Rightarrow \frac{S}{2} = \frac{a}{1-r} = \frac{1/2}{1-1/2} = \frac{1/2}{1/2}$

$\therefore S = 2$

$\therefore E(X) = 2$

Thus, the expectation of no. of tosses required is 2.

Ans

$E(X) = \lambda = np$ | $P(X=x) = \frac{(e^{-\lambda}) \lambda^x}{x!}$; $x = 0, 1, 2, \ldots \infty$

$V(X) = \lambda = np$ | $SD = \sigma = \sqrt{np}$; $\lambda = np$

Poisson Distribution

1) If average number of customers arrived at a certain bank per hours is 2, find the probability that

i> No. of customer per hours?

ii) At least 3 customers per hour?

iii> At most 1 customer per hour?

iv> Exactly 5 customers in 2 hours?

Soln

Let X be a random variables that denotes the no. of customers arrived ~~bank~~ at bank per hour

i.e. $X = 0, 1, 2, \ldots \infty$.

Average, $\lambda = 2$

we have,

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} ; x = 0, 1, 2, \ldots$$

$$= \frac{e^{-2} \cdot 2^x}{x!}$$

i> Prob (no customer per hour): $P(X=0)$

$$= \frac{e^{-2} 2^0}{0!} = 0.135 \text{ Ans}$$

ii> Prob(at least 3 customers per hour): $P(X \geq 3) = 1 - P(X<3)$

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - \left(\frac{e^{-2} 2^0}{0!} + \frac{e^{-2} \cdot 2^1}{1!} + \frac{e^{-2} \cdot 2^2}{2!} \right)$$

$$= 0.323 \text{ Ans}$$

iii> Prob(at most 1 customer per hour) : $P(X \leq 1) =$

$= P(X=0) + P(X=1)$

$= \frac{e^{-2} \cdot 2^0}{0!} + \frac{e^{-2} \cdot 2^1}{1!}$

$=$ ~~0.4 06~~ 0.406 Ans

iv> Let X be the random variable that denotes no. of customer in 2 hours.

$\therefore \lambda = 2\lambda = 2 \times 2 = 4$

$\therefore$ Prob(exactly 5 customers in 2 hours) $= P(X=5)$

$= \frac{e^{-4} 4^5}{5!} = 0.156$ Ans

(2) If 5% of the electric bulbs manufactured by a company are defective. Use poisson distribution to find the probability that in a sample of 100 bulbs,

i> none is defective

ii> at least 1 is defective

iii> at most 2 bulbs are defective.

iv> 5 bulbs are defective.

Sol

Let X be a random variable that denotes the no. of defective bulbs.

i.e. $X = 0, 1, 2, 3, \ldots \ldots 100$

no. of sample, $n = 100$

prob of bulbs, $p = 5\% = 0.05$

Average occurance, $\lambda = np = 100 \times 0.05 = 5$

we have,

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-5} \cdot 5^x}{x!}$$

i> Prob(none is defective) $= P(X=0)$

$$= \frac{e^{-5} \cdot 5^0}{0!} = 0.0067 \text{ Ans}$$

ii> Prob(at least 1 is defective): $P(X \geq 1) = 1 - P(X<1)$

$$= \cancel{P(X=0)} \; 1 - P(X=0)$$

$$= 1 - 0.0067$$

$$= 0.9933 \text{ Ans}$$

iii> Prob(at most 2 bulbs are defective): $P(X \leq 2)$

$$= P(X=0) + P(X=1) + P(X=2)$$

$$= \frac{e^{-5} 5^0}{0!} + \frac{e^{-5} 5^1}{1!} + \frac{e^{-5} 5^2}{2!} = e^{-5}\left(1 + 5 + \frac{25}{2}\right)$$

$$= 0.0067 \times 18.5$$

$$= 0.12395 \text{ Ans}$$

iv> Prob(5 bulbs are defective): $P(X=5) = \frac{e^{-5} 5^5}{5!} = 0.175$ Ans

Pre-Board.
2022-Fall

3.a> The number of a accident in a year attributed to taxi driver in a city is Poisson distribution with mean 3. Out of 1000 taxi driver, find approximately the number of driver with

i> more than 3 accidents in a year,

ii> less than 2 accident.

Sol

Let X be a random variable that denotes the no. of accidents in a year.

i.e. X is 0, 1, 2, 3, ...

we have, $\lambda = 3, n = 1000$

$$P(X=x) = \frac{e^{-3} 3^x}{x!}$$

i> P(more than 3 accidents in a year) : $P(X>3)$

$= 1 - P(X \leq 3)$

$= 1 - P(X=0) - P(X=1) - P(X=2) - P(X=3)$

$= 1 - \frac{e^{-3} 3^0}{0!} - \frac{e^{-3} 3^1}{1!} - \frac{e^{-3} 3^2}{2!} - \frac{e^{-3} 3^3}{3!}$

$= 1 - e^{-3}\left(1 + 3 + \frac{9}{2} + \frac{27}{6}\right)$

$= 1 - e^{-3} \times 13$

$= 0.35$ Ans

ii) P(less than 2 accident) = $P(X<2)$

$$= P(X=0) + P(X=1)$$

$$= \frac{e^{-3}3^0}{0!} + \frac{e^{-3}3^1}{1!}$$

$$= e^{-3}(1+3)$$

$$= 4e^{-3}$$

$$= 0.199$$ Ans

(3) . A car hire company has 2 cars which it hires out day by day. The number of demands for a car on each day is distributed as poisson distribution with mean 1.5. Calculate proportion days on which

i) neither car is used.

ii) Some demand is refused.

Sol: Here

Let x be a random variable that denotes the no. of demands in a day.

i.e. $X = 0, 1, 2, 3, \ldots \infty$

Average or mean, $\lambda = 1.5$

i) Prob(neither car is used): $P(X=0)$

$= \frac{e^{-1.5} 1.5^0}{0!} = 0.2231$ Ans

ii) Prob(some demand is refused):

$P(X>2) = 1 - P(X \leq 2)$

$= 1 - P(X=0) - P(X=1) - P(X=2)$

$= 1 - \left(\frac{e^{-1.5} 1.5^0}{0!} + \frac{e^{-1.5} 1.5^1}{1!} + \frac{e^{-1.5} 1.5^2}{2!} \right)$

$= 1 - e^{-1.5} \left(1 + 1.5 + \frac{2.25}{2} \right)$

$= 0.191$ Ans

Gamma Distribution:

$$f(x) = \frac{1}{\beta^{\alpha}\Gamma\alpha} x^{\alpha-1} e^{-x/\beta} \quad ; \quad 0 \leq x \leq \infty$$

$E(X) = \alpha\beta$; $V(X) = \alpha\beta^2$; S.D.(σ) = $\beta\sqrt{\alpha}$

$\Gamma n = (n-1)!$

$\int_0^\infty e^{-ax} x^{n-1} dx = \frac{\Gamma n}{a^n}$

① In a certain city, daily consumption of water (in millions of gallons) follow approximately ~~α=2 & β=3 If the~~ a Gamma distribution with parameter $\alpha = 2$ & $\beta = 3$. If the daily capacity of this city is 9 million gallon of water, what is the probability that on any given day the water supply is inadequate?

Sol:

Let X be a random variable that denotes the daily consumption of water

Here, $\alpha = 2$ & $\beta = 3$

The p.d.f of gamma function is given by

$$f(x) = \frac{1}{\beta^{\alpha}\Gamma\alpha} \cdot x^{\alpha-1} e^{-x/\beta}$$

$$= \frac{1}{3^2\Gamma 2} \cdot x^{2-1} e^{-x/3}$$

$$= \frac{1}{9\Gamma 2} x e^{-x/3}$$

$$= \frac{1}{9(2-1)!} x e^{-x/3}$$

$$\therefore f(x) = \frac{1}{9} x e^{-x/3}$$

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Prob (that water supply is inadequate on given day):

$P(X>9) = 1 - P(X \le 9)$

$= 1 - \int_0^9 f(x)\, dx$

$= 1 - \int_0^9 \frac{1}{9} x e^{-x/3}\, dx$

$\therefore P(X>9) = 0.199$ Ans

②. In a certain city, the daily consumption of electric power (in millions of kwh) can be treated as a random variable having a gamma distribution with $\alpha=3$ & $\beta=2$. If the power plant of this city has a daily capacity of 12 millions kwh. What is the probability that this power supply will be inadequate on any given day?

Sol:

Let X be a random variable that denotes the daily consumption of electric power.

Here,

$\alpha = 3$ & $\beta = 2$

The p.d.f. of gamma function is given by

$$f(x) = \frac{1}{\beta^\alpha \Gamma\alpha} x^{\alpha-1} e^{-x/\beta}$$

CamScanner

$$= \frac{1}{2^3 \Gamma 3} x^{3-1} e^{-x/2}$$

$$\therefore f(x) = \frac{1}{8(3-1)!} x^2 e^{-x/2} \quad (\because \Gamma n = (n-1)!)$$

$$\therefore f(x) = \frac{1}{16} x^2 e^{-x/2}$$

Prob (power supply will be inadequate on any given day)

$= P(X > 12)$

$= \cancel{P} \; 1 - P(X \leq 12)$

$= 1 - \int_0^{12} f(x)\, dx$

$= 1 - \int_0^{12} \frac{1}{16} x^2 e^{-x/2}$

$\therefore = P(X > 12) = 0.0619$ Ans

2022-Fall
Pre-Board

6.a) Suppose that the life time of a certain kind of Computer is random variable x having the Gamma distribution with parameter $\alpha=2$ and $\beta=3$, if the daily capacity of this city is 9 million gallon of water, what is the probability that on any given day the water supply is inadequate?

Sol

Let x be a random variable that denotes the daily consumption of water,

Here, $\alpha=2$ & $\beta=3$

The p.d.f of gamma function is given by

$$f(x) = \frac{1}{\beta^{\alpha}\,\Gamma\alpha}\, x^{\alpha-1} e^{-x/\beta}$$

$$= \frac{1}{3^{2}\,\Gamma 2}\, x^{2-1} e^{-x/3}$$

$$= \frac{1}{9\,(2-1)!}\, x e^{-x/3}$$

$$\therefore f(x) = \frac{x e^{-x/3}}{9}$$

Prob(water supply is inadequate on any given data):

$P(X > 9) = 1 - P(X \le 9)$

$= 1 - \int_0^9 f(x)\,dx$

$= 1 - \int_0^9 \frac{x e^{-x/3}}{9}\,dx$

$= 0.199$ Ans

$E(X) = \frac{\alpha}{\alpha+\beta}$ Beta distribution

$$f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

$$B(\alpha,\beta) = \frac{\Gamma\alpha \, \Gamma\beta}{\Gamma(\alpha+\beta)}$$

$$= B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$$

$$\cancel{B(\alpha+\beta)} = \frac{\Gamma\alpha \, \Gamma\beta}{\Gamma(\alpha+\beta)}$$

1. In a certain hill, the proportion of soil lapsed in 10 years is a random variable having beta distribution with $\alpha = 3$ and $\beta = 2$.

i> Find the average what percent of soil lapse in a given 10 years?

ii> Find the probability that at most 25% of soil will lapse in any given 10 years.

Sol

Let x be a random variable that denotes proportion of soil lapse in 10 years.

Here, $\alpha = 3$, $\beta = 2$

i> Mean of beta distribution $= \frac{\alpha}{\alpha+\beta} = \frac{3}{3+2} = 0.6$ Ans

$\therefore$ The percent of soil lapse $= 0.6 \times 100\% = 60\%$ in 10 years. Ans

ii> The probability density function, p.d.f of beta distribution is given by

$$f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1} = \frac{1}{\frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\alpha+\beta)}} x^{3-1}(1-x)^{2-1}$$

$$= \frac{x^2(1-x)}{\frac{\Gamma 3\,\Gamma 2}{\Gamma 5}} = \frac{x^2(1-x)\,(5-1)!}{(3-1)!\,(2-1)!} = \frac{24x^2(1-x)}{2}$$

$\therefore f(x) = 12x^2(1-x)$

Now,

Prob(at most 25% of soil will lapse in given 10 years) =

$= P(X \le 0.25)$

$= \int_0^{0.25} f(x)\,dx$

$= \int_0^{0.25} 12x^2(1-x)\,dx$

$= 0.5078$ Ans

Q2. Verify for $\alpha = 3$ & $\beta = 3$ that the integral of the beta density, from 0 to 1, is equal to 1.

Sol Let X be a random variable that follows beta distribution with $\alpha = 3$ & $\beta = 3$

The p.d.f. of beta distribution is given by

$f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}$

$= \frac{1}{\frac{\Gamma\alpha\,\Gamma\beta}{\Gamma(\alpha+\beta)}} x^{3-1}(1-x)^{3-1} = \frac{\Gamma 3+3}{\Gamma 3\,\Gamma 3} x^2(1-x)^2$

$= \frac{(6-1)!\,x^2(1-x)^2}{2!\,2!} = \frac{\cancel{120}^{30}}{\cancel{4}} x^2(1-x^2)$

$\therefore f(x) = 30x^2(1-x^2)$

(3) Now, Total integration:

$$\int_0^1 f(x)\,dx = \int_0^1 30x^2(1-x)^2\,dx$$

$$= 1 \quad \text{Ans}$$

verified.

Numericals

1) In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution.

Sol

Let X be a random variable such that $X \sim N(\mu, \sigma^2)$.

Given;

$P(X < 45) = 31\% = 0.31$

$P(X > 64) = 8\% = 0.08$

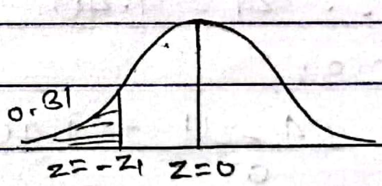
$\mu = ?$; $\sigma = ?$

we know,

standard normal variate, $z = \frac{X-\mu}{\sigma}$

* when $X = 45$,

$$z = \frac{45-\mu}{\sigma} = -z_1 \text{ (say)} \quad \text{———} (1)$$



Now,

$P(z < -z_1) = 0.31$

$\Rightarrow 0.5 - P(-z_1 < z < 0) = 0.31$

$\Rightarrow 0.5 - 0.31 = P(-z_1 < z < 0)$

$\Rightarrow 0.19 = P(0 < z < z_1)$

$\Rightarrow 0.19 = P(0 < z < z_1)$

$\Rightarrow P(0 < z < z_1) = 0.19$

$\Rightarrow P(0 < z < 0.50) = 0.19$ (From Normal table)

$\therefore z_1 = 0.50$

From (1),

$$\frac{45-\mu}{\sigma} = -0.5$$

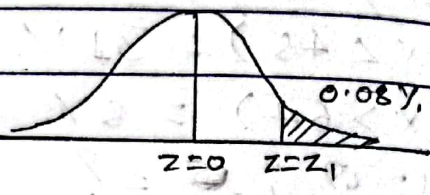
$\therefore 45 - \mu = -0.5\sigma$ ——— (2)

(3) Again, when $X = 64$,

$$Z = \frac{64-\mu}{\sigma} = z_1 \text{ (say)} \quad \text{——— (3)}$$

$P(X > 64) = 0.08$

$\Rightarrow P\left(\frac{X-\mu}{\sigma} > \frac{64-\mu}{\sigma}\right) = 0.08$



$\Rightarrow P(Z > z_1) = 0.08$

$\Rightarrow 0.5 - P(Z < z_1) = 0.08$

$\Rightarrow 0.5 - P(0 < Z < z_1) = 0.08$

$\Rightarrow 0.5 - 0.08 = P(0 < Z < z_1)$

$\Rightarrow P(0 < Z < z_1) = 0.42$

$\Rightarrow P(0 < Z < 1.40) = 0.42$

$\therefore z_1 = 1.40$

From (3),

$$\frac{64-\mu}{\sigma} = 1.40$$

$\therefore 64 - \mu = 1.40\sigma$ ——— (4)

From (2) & (4), we get:-

$\sigma = 10$ <u>Ans</u>

$\mu = 50$ <u>Ans</u>

② The average income and variance of 10,000 workers per day are \$70 and \$25 respectively. Find the number of worker whose

i> income is less than 74.

ii> more than 76.

iii> between 65 and 78.

Sol. Given:-

No. of worker, $n = 10{,}000$

Average income, $\mu = \$70$

variance, $\sigma^2 = \$25$

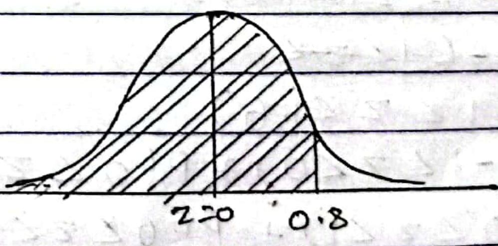
$S.D. = \sqrt{\sigma^2} = \sqrt{25} = 5$

Let X be a random variable that denotes the income of 10,000 workers.

i> $P(X<74)$:

$P\left(\frac{X-\mu}{\sigma} < \frac{74-\mu}{\sigma}\right)$

$= P\left(Z < \frac{74-70}{5}\right)$



$= P(Z < 0.8)$

$= \cancel{P(0<Z<0.8)}$

$= 0.5 + P(0 < Z < 0.8)$

$= 0.5 + 0.28814$

$= 0.78814$

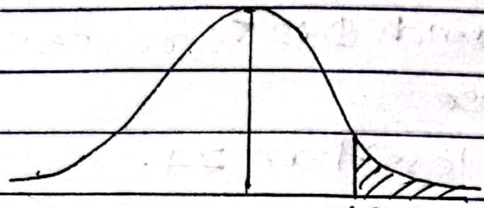
$\therefore P(X<74) = 0.78814$

$\therefore$ The no. of worker whose income is less than \$74 is

$= 10000 \times 0.78814 = 7881.4 \approx 7881$ Ans

(3) ii> $P(X > 76)$

$= P\left(\frac{X-\mu}{6} > \frac{76-70}{5}\right)$



Z=0 Z=1.2

$= P(Z > 1.2)$

$= 0.5 - P(0 < Z < 1.2)$

$= 0.5 - 0.38493$ (From Normal table)

$= 0.11507$

$\therefore P(X > 76) = 0.11507$

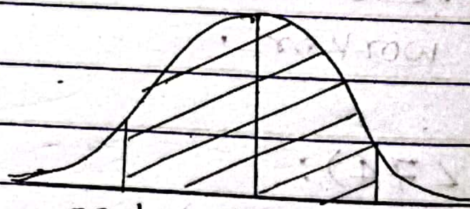
$\therefore$ The no. of worker whose income is more than \$76

$= 0.11507 \times 10,000$

$= \$\ 1150.7$ Ans

iii> $P(65 < X < 78)$:

$= P\left(\frac{65-70}{5} < \frac{X-\mu}{6} < \frac{78-70}{5}\right)$



z=-1 z=0 z=1.6

~~$= P(-1 < z < 1.6)$~~

~~$= 0.5 - (-1 < z <$~~

$= P(-1 < z < 1.6)$

$= P(-1 < z < 0) + P(0 < z < 1.6)$

$= P(0 < z < 1) + P(0 < z < 1.6)$ (by symmetry)

$= 0.34134 + 0.4452$

$\therefore P(65 < X < 78) = \$0.78654$

$\therefore$ The no. of worker whose income is between 65 and 78\$

$= 10000 \times 0.78654 = 7865.4$ Ans

③ Given a normal distribution with mean, $\mu = 100$ and S.D, $\sigma = 10$. What is the probability that

a> $X > 75$?

b> $X \geq 70$?

c> $75 < X < 85$?

d> $X < 80$ or $X > 110$?

e> 70% of the values are less than what values of x?

f> 70% of the values are above what x value?

g> 80% of the values are between what two x values?

Sol Given:

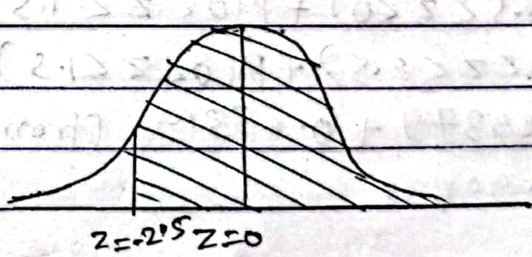
$\mu = 100$

$\sigma = 10$

Let X be a random variable that denotes $X \sim N(\mu, \sigma^2)$.

i> $P(X > 75)$:

$= P\left(\frac{X-\mu}{\sigma} > \frac{75-100}{10}\right)$



$= P(z > -2.5)$

$= 0.5 + P(-2.5 < z < 0)$

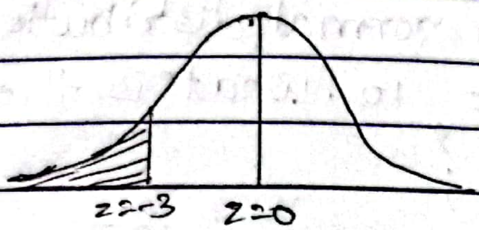
$= 0.5 + P(0 < z < 2.5)$ (By symmetry)

$= 0.5 + 0.49379$ (From Normal table)

$= 0.99379$ Ans

(3) b) $P(X < 70)$:

$= P\left(\frac{X-\mu}{\sigma} < \frac{70-100}{10}\right)$



$= P(z < -3)$

$= 0.5 - P(-3 < z < 0)$

$= 0.5 - P(0 < z < 3)$

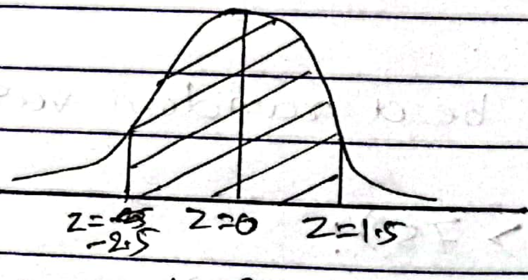
$= 0.5 - 0.49865$ (From Normal table)

$= 0.00135$

$\therefore P(X < 70) = 0.00135$ Ans

c) $P(75 < X < 115)$

$= P\left(\frac{75-100}{10} < \frac{X-\mu}{\sigma} < \frac{115-100}{10}\right)$



$= P(-2.5 < z < 1.5)$

$= P(-2.5 < z < 0) + P(0 < z < 1.5)$

$= P(0 < z < 2.5) + P(0 < z < 1.5)$ (By symmetry)

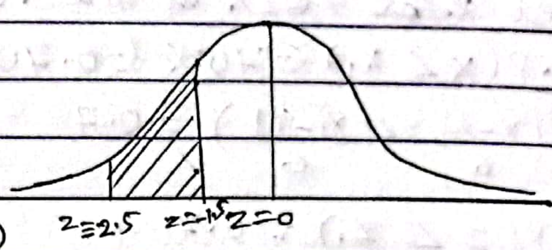
$= 0.49379 + 0.43319$ (from Normal table)

$= 0.92698$

$\therefore P(75 < X < 85) = 0.92698$ Ans

d) $P(75 < X < 85)$:

$= P\left(\frac{75-100}{10} < \frac{X-\mu}{\sigma} < \frac{85-100}{10}\right)$



$= P(-2.5 < Z < -1.5)$

$= P(-2.5 < Z < 0) - P(-1.5 < Z < 0)$

$= P(0 < Z < 2.5) - P(0 < Z < 1.5)$ (By Symmetry)

$= 0.49379 - 0.43319$ (From Normal table)

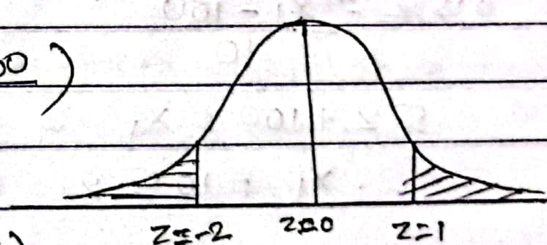
$= 0.0606$

$\therefore P(75 < X < 85) = 0.0606$ Ans

e) $P(X < 80)$ or $P(X > 110)$:

$= \cancel{P\left(\frac{80-100}{10}\right.}$

$P\left(\frac{X-\mu}{\sigma} < \frac{80-100}{10}\right) + P\left(\frac{X-\mu}{\sigma} > \frac{110-100}{10}\right)$



$= P(Z < -2) + P(Z > 1)$

$= 0.5 - P(-2 < Z < 0) + 0.5 - P(0 < Z < 1)$

$= 1 - P(0 < Z < 2) - P(0 < Z < 1)$ (By symmetry)

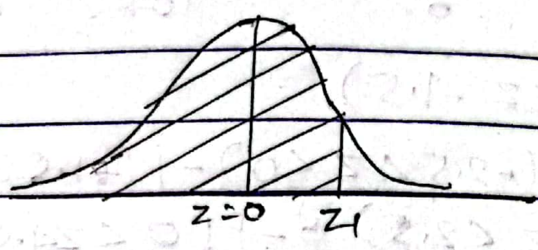
$= 1 - 0.47725 - 0.34134$ (From Normal table)

$= 0.18141$

$\therefore P(X < 80$ or $X > 110) = 0.18141$ Ans f> Let ~~x~~ 70 % of the values are less than x_1.

i.e. $P(x < x_1) = 70\% = 0.70$

$\Rightarrow P\left(\frac{x-\mu}{\sigma} < \frac{x_1-\mu}{\sigma}\right) = 0.7$



$\Rightarrow P(z < z_1) = 0.7$

$\Rightarrow 0.5 + P(0 < z < z_1) = 0.7$

$\Rightarrow P(0 < z < z_1) = 0.7 - 0.5$

$\Rightarrow P(0 < z < z_1) = 0.2000$

$\Rightarrow P(0 < z < 0.52) = 0.2000$ (From Normal table)

$\therefore z_1 = 0.52$

Now,

$z_1 = \frac{x_1 - \mu}{\sigma}$

$\Rightarrow 0.52 = \frac{x_1 - 100}{10}$

$\Rightarrow 5.2 + 100 = x_1$

$\therefore x_1 = 105.2$ Ans

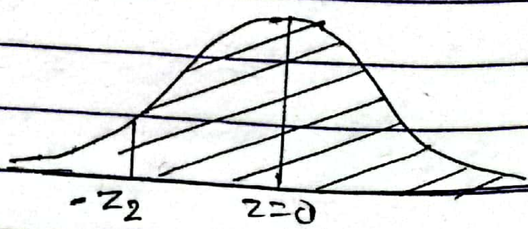
$\therefore$ The 70 % of the values are less than 105.2.

g> Let 70 % of the values will be above x_2.

i.e. $P(x_2 < x) = 70\% = 0.70$

$\Rightarrow$ ~~P(z2 < z)~~

$\Rightarrow P\left(\frac{x_2-\mu}{\sigma} < \frac{x-\mu}{\sigma}\right) = 0.70$



$\Rightarrow P(-z_2 < z) = 0.70$

$P(-z_2 < z) = 0.70$

$\Rightarrow \cancel{P(-z_2 < z < 0)}\ 0.5 + P(-z_2 < z < 0) = 0.70$

$\Rightarrow 0.5 + P(0 < z < z_2) = 0.70$

$\Rightarrow P(0 < z < z_2) = 0.2$

$\Rightarrow P(0 < z < 0.52) = 0.2000$

$\therefore z_2 = 0.52$

$\because -z_2 = \dfrac{x_2 - \mu}{\sigma}$

$\Rightarrow -0.52 = \dfrac{x_2 - 100}{10}$

$\Rightarrow -5.2 + 100 = x_2$

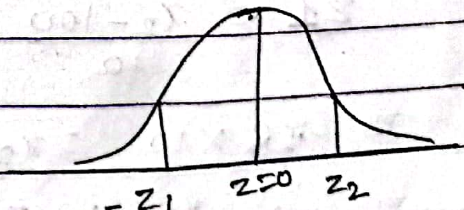
$\therefore x_2 = 94.8$

$\therefore$ The 70% of the values are above 94.8 Ans.

h) Let 80% of the values are between x_1 and x_2.

i.e. $P(x_1 < X < x_2) = 80\% = 0.80$

$\Rightarrow P\left(\dfrac{x_1 - \mu}{\sigma} < \dfrac{X - \mu}{\sigma} < \dfrac{x_2 - \mu}{\sigma}\right) = 0.8$



$\Rightarrow P(-z_1 < Z < z_1) = 0.8$

where,

$-z_1 = \dfrac{x_1 - 100}{10}$ — (1)

$z_1 = \dfrac{x_2 - 100}{10}$ — (2)

$\Rightarrow P(-z_1 < z < z_1) = 0.80$

$\Rightarrow P(-z_1 < z < 0) + P(0 < z < z_1) = 0.80$

$\Rightarrow P(0 < z < z_1) + P(0 < z < z_1) = 0.80$ (By symmetry)

$\Rightarrow 2P(0 < z < z_1) = 0.80$

$\Rightarrow P(0 < z < z_1) = 0.4$

$\Rightarrow P(0 < z < 1.28) = 0.4$

$\therefore z_1 = 1.28$

from (I),

$$-z_1 = \frac{x_1 - 100}{100}$$

$$\Rightarrow -1.28 = \frac{x_1 - 100}{10}$$

$\Rightarrow -12.8 + 100 = x_1$

$\therefore x_1 = 87.2$

Also from (II),

$$z_1 = \frac{x_2 - 100}{10}$$

$\Rightarrow 1.28 \times 10 = x_2 - 100$

$\Rightarrow \therefore x_2 = 12.8 + 100$

$\therefore x_2 = 112.8$

$\therefore$ The 80% of the values lies between 87.2 and 112.8. Ans

5. In a certain examination test, 2000 students appeared in statistics. The average marks obtained by 2000 student was 50 and standard deviation was 5.

i) How many students do you expect to obtain more than 60% marks?

ii) What are the maximum marks of the top 100 students.

iii) What is the maximum marks below 200 students?

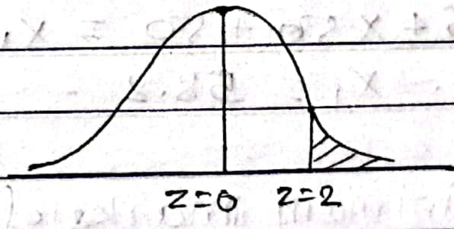
Assume that the marks are normally distributed.

Sol

Let X be a random variable that denotes the no. of students appeared in statistics examination such that $X \sim N(50, 25)$.

i) $P(X>60)$:

$P\left(\frac{X-\mu}{\sigma} > \frac{60-50}{5}\right)$



$= P(Z>2)$

$= 0.5 - P(0<Z<2)$

$= 0.5 - 0.47725$ (From Normal table)

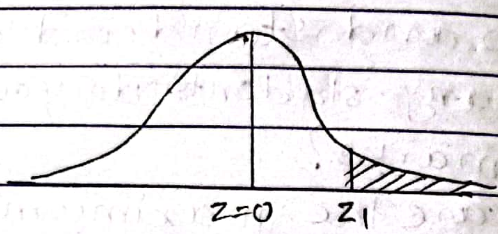
$\therefore P(X>60) = 0.02275$

$\therefore$ The no. of students we expect to obtain ~~60%~~ more than 60% marks $= 2000 \times 0.02275 = 45.5 = 46$ Ans.

(3) ii) Let x_1 be the minimum marks of top 100 students,

i.e. $P(X > x_1) = \frac{100}{2000} = 0.05$

$\Rightarrow P\left(\frac{X-\mu}{\sigma} > \frac{x_1-\mu}{\sigma}\right) = 0.05$



$\Rightarrow P(Z > Z_1) = 0.05$

$\Rightarrow 0.5 - P(0 < Z < Z_1) = 0.05$

$\Rightarrow 0.5 - 0.05 = P(0 < Z < Z_1)$

$\Rightarrow P(0 < Z < Z_1) = 0.45$

$\Rightarrow P(0 < Z < 1.64) = 0.45$ (From Normal table)

$\therefore Z_1 = 1.64$

$\because Z_1 = \frac{x_1-\mu}{\sigma} = \frac{x_1 - 50}{50}$

$\Rightarrow 1.64 \times 50 + 50 = x_1$

$\therefore x_1 = 58.2$

$\therefore$ The minimum marks of top 100 students is 58.2 Ans

iii) Let x_1 be the minimum marks obtained below 200 students

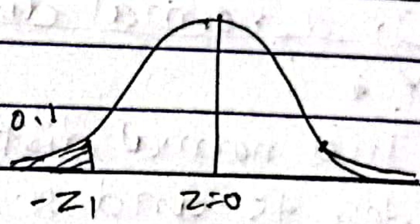
then, ~~$P(X<200)$~~ =

~~$P(X>x_1)$~~

$P(X < x_1) = \frac{200}{2000} = 0.1$

$\Rightarrow P\left(\frac{X-\mu}{\sigma} < \frac{x_1-\mu}{\sigma}\right) = 0.1$

$\Rightarrow P(Z < -z_1) = 0.1$



$\Rightarrow \cancel{0.5 - P(0 < z < -z_1)} = 0.1$

$\Rightarrow \cancel{0.5 - 0.1 = P(0 < z_1 < z)}$

$\Rightarrow 0.5 - P(-z_1 < z < 0) = 0.1$

$\Rightarrow 0.5 - 0.1 = P(-z_1 < z < 0)$

$\Rightarrow 0.4 = P(0 < z < z_1)$ (By symmetry)

$\Rightarrow P(0 < z < z_1) = 0.4$

$\Rightarrow P(0 < z < 1.28) = 0.8$ (From Normal table)

$\therefore z_1 = 1.28$

Now,

$$\frac{x_1 - \mu}{\sigma} = -z_1$$

$$\Rightarrow \frac{x_1 - 50}{5} = -1.28$$

$\Rightarrow x_1 - 50 = -1.28 \times 5$

$\Rightarrow x_1 = 50 + 1.28 \times 5$

$\therefore x_1 = 56.4$

$\therefore$ The minimum marks obtained below 200 students is 56.4 %. Ans

Pre-Board
2022-Fall

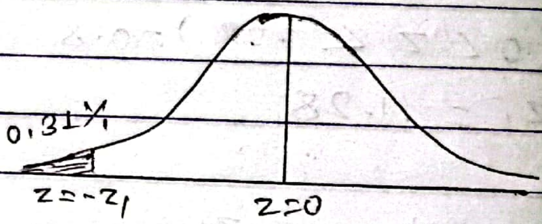
3.b> In a normal ~~distribution of random variables X and Y~~ i

3.b> In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and S.D. of the distribution.

Sol

Let X be a random variable that denotes a normal distribution such that $X \sim N(\mu, \sigma^2)$.

i> when $(X < 45)$.



$P(X > 45) = 31\%$

$\Rightarrow P\left(\frac{X-\mu}{\sigma} > \frac{45-\mu}{\sigma}\right) = 0.31$

$\Rightarrow P(Z > -z_1) = 0.31$

$\Rightarrow 0.5 - P(-z_1 < Z < 0) = 0.31$

$\Rightarrow 0.5 - 0.31 = P(0 < Z < z_1)$

$\Rightarrow 0.19 = P(0 < Z < z_1)$

$\Rightarrow P(0 < Z < z_1) = 0.19$

$\Rightarrow P(0 < Z < 0.5) = 0.19$ (From Normal table)

$\therefore z_1 = 0.5$

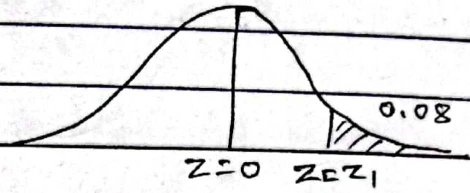
$\because \frac{45-\mu}{\sigma} = -0.5$

$\Rightarrow 45 - \mu = -0.5\sigma$

$\therefore -0.5\sigma + \mu = 45$ ——(1)

ii) when $(X < 64)$:

i.e. $P(X < 64) = 8\%$



$$\Rightarrow P\left(\frac{X-\mu}{\sigma} < \frac{64-\mu}{\sigma}\right) = 8\%$$

$$\Rightarrow P(Z < Z_1) = 0.08$$

$$\Rightarrow 0.5 - P(0 < Z < Z_1) = 0.08$$

$$\Rightarrow 0.5 - 0.08 = P(0 < Z < Z_1)$$

$$\Rightarrow P(0 < Z < Z_1) = 0.42$$

$$\Rightarrow P(0 < Z < 1.40) = 0.42$$

$$\therefore Z_1 = 1.40$$

Now,

$$\frac{64-\mu}{\sigma} = 1.4$$

$$\Rightarrow 64 - \mu = 1.4\sigma$$

$$\therefore 1.4\sigma + \mu = 64 \quad \text{———} \quad \text{(ii)}$$

From (i) & (ii), we get:-

$\sigma = 10$

$\mu = 50$ Ans